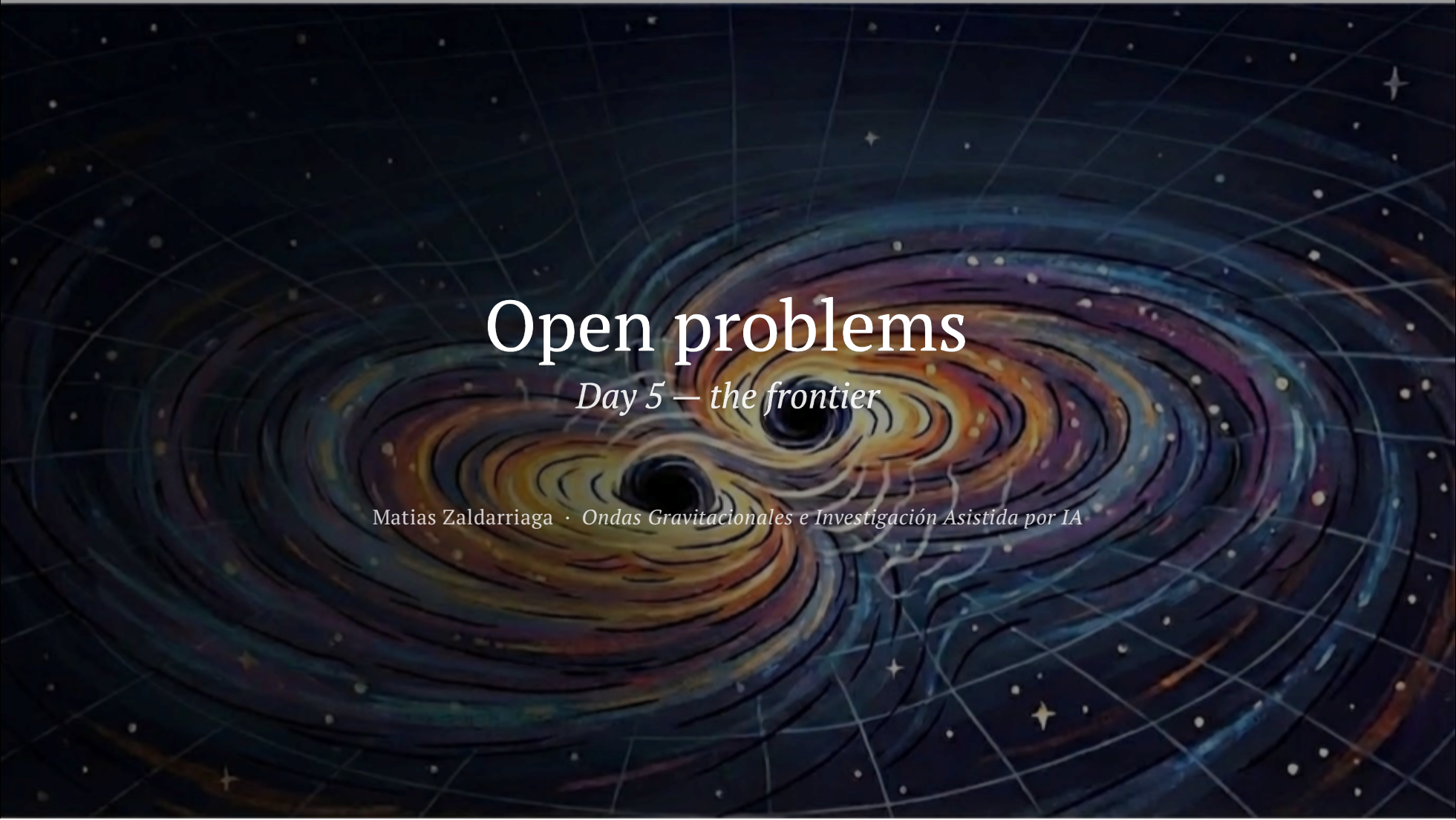


The background of the slide is a dark blue space filled with a grid of white lines representing spacetime. Two large, swirling, colorful patterns (yellow, orange, and blue) represent gravitational waves. These waves are moving towards the center and then splitting into two smaller waves. The overall effect is a dynamic, swirling motion. The text is centered over this background.

Open problems

Day 5 — the frontier

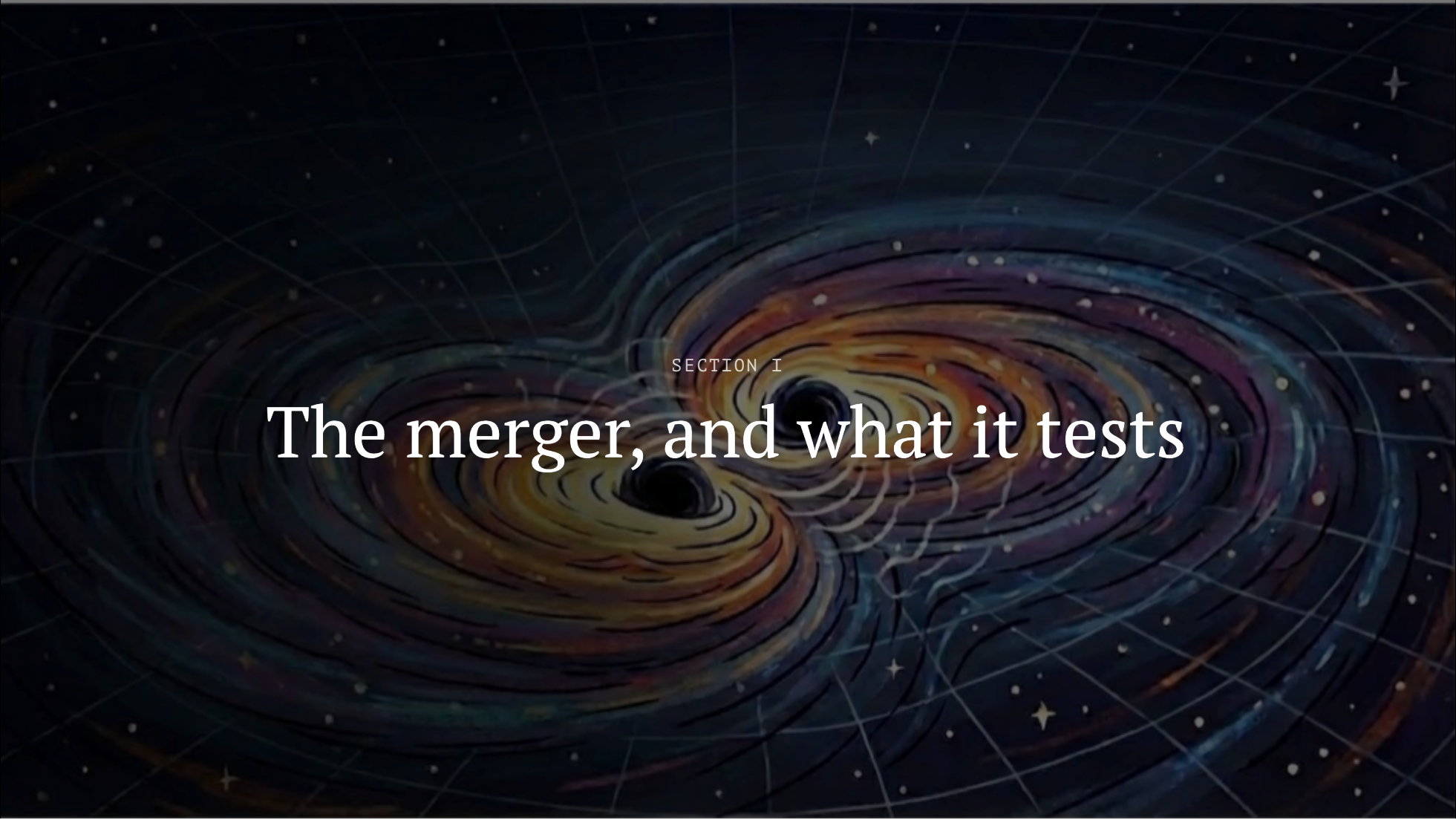
Matias Zaldarriaga · *Ondas Gravitacionales e Investigación Asistida por IA*

Four things the week *opened*.

Days 1 to 4 each ran one confrontation between theory and data to the end. Today does not run one. It says where the ends are.

1. **The merger.** Day 2 said out loud that an inspiral template cannot reach it. What happens instead, and what it tests.
2. **GW170817.** One event, several sciences — and the tightest connection in this course between gravitational waves and cosmology.
3. **Louder and lower.** What a factor of ten in sensitivity buys, and what the millihertz band has that this one does not.
4. **What is unsettled in our own work.** The heaviest event, the nanohertz amplitude, lensing.

Each of the four ends on a problem, by name. The problems are collected in one place and you leave with it.

The background of the slide is a dark, deep blue space filled with a grid of thin, light blue lines representing a gravitational well or spacetime curvature. Two large, swirling galaxies are the central focus, their spiral arms rendered in vibrant shades of orange, yellow, and blue. They are positioned close together, suggesting a gravitational interaction or merger. Scattered throughout the scene are numerous small, white, four-pointed stars of varying sizes.

SECTION I

The merger, and what it tests

After the *inspiral*.

The post-Newtonian expansion is an expansion in v/c , and at the last stable orbit v/c is already about a third. Day 2 measured what that costs: a template that stops at the last stable orbit collects less than half the available signal-to-noise for a heavy binary.

What happens instead is not a small correction to the inspiral. Two horizons merge into one, and what is left is a single distorted black hole that has to shed its distortion before it can be stationary.

THREE REGIMES, ONE SIGNAL

Inspiral — two bodies, post-Newtonian, analytic.

Merger — strong field, no expansion parameter, numerical relativity only.

Ringdown — one body, perturbed, analytic again.

The middle one is the reason waveform models are fitted to simulations rather than written down.

Quasi-normal *modes*.

The perturbed remnant rings like a struck bell, and the ringing is a sum of damped sinusoids:

$$h(t) \sim \sum_{\ell mn} A_{\ell mn} e^{-t/\tau_{\ell mn}} \cos(2\pi f_{\ell mn} t + \phi_{\ell mn})$$

The amplitudes A and phases ϕ depend on how the hole was made. The frequencies and damping times do **not**. Each $f_{\ell mn}$ and $\tau_{\ell mn}$ is fixed by the remnant's mass and spin, and by nothing else — which is the no-hair statement in a form you can measure.

Dimensionally there is only one time in the problem, GM/c^3 , so $f \propto 1/M$ and τ is a few of those. For a sixty-solar-mass remnant that is a couple of hundred hertz and a few milliseconds — a handful of cycles. The whole test lives in those cycles.

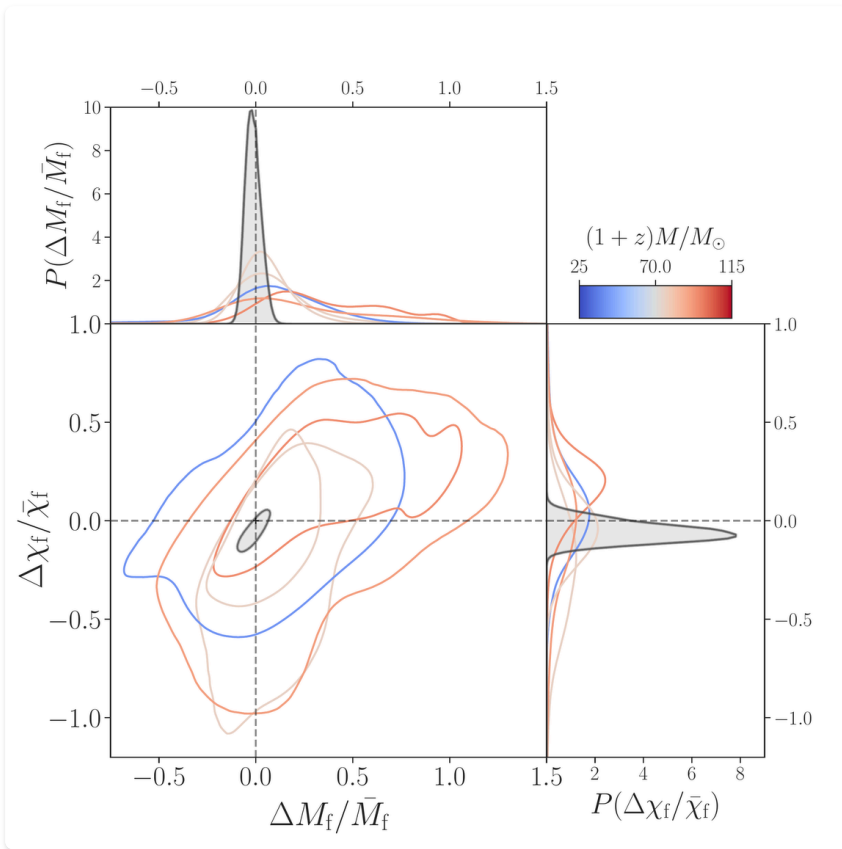
The remnant, measured *twice*.

The inspiral fixes the two component masses and spins. Feed those through a numerical-relativity fit and you have a **prediction** for the remnant's mass and spin.

The part after the merger **measures** the same two numbers, from the frequencies and damping times directly.

Two routes to one pair of numbers. Plot the fractional difference of each, and general relativity says the contours contain the origin. They do.

The sharper version of the same idea uses two ringdown modes and no inspiral at all: two modes are four numbers, and the remnant has two. Everything past the second is a test. That needs signal-to-noise we mostly do not have yet.



Borrowed: LIGO-Virgo-KAGRA, Tests of general relativity with GWTC-3

(arXiv:2112.06861) Fig. 6. Grey: all events combined

A visualization of gravitational waves, showing two black holes in the process of merging. The background is a dark blue space with a grid of white lines representing spacetime. Concentric, swirling bands of color (yellow, orange, red, and purple) represent the ripples in spacetime. The two black holes are depicted as dark, circular regions with bright, glowing centers, positioned close together in the center of the frame. The text "SECTION II" is centered above the main title.

SECTION II

GW170817

One event, several *sciences*.

Two neutron stars at about 40 Mpc. The inspiral was in band for roughly a minute, because the masses are small and the chirp is slow.

THE SEQUENCE

- the merger, in the ground-based band;
- a short gamma-ray burst, 1.74 s later;
- an optical counterpart, located within a day;
- a kilonova, reddening over a fortnight.

WHAT CAME OUT OF IT

- the speed of gravitational waves;
- how much the stars deform, and so what matter does above nuclear density;
- H_0 , with no distance ladder;
- where the heavy elements are made.

Four unrelated measurements from one event. Nothing else in this field has that property, and the reason is the counterpart.

The speed of *propagation*.

Gravitational waves and photons left the same place at very nearly the same time and arrived 1.74 s apart, over a light travel time of about 10^8 years. The only quantity not measured is when the gamma rays were emitted; assume zero to ten seconds after the merger.

$$\frac{\Delta v}{v} \simeq \frac{c \Delta t}{D} \implies -3 \times 10^{-15} < \frac{\Delta v}{v} < +7 \times 10^{-16}, \quad \Delta v \equiv v_{\text{GW}} - v_{\text{EM}}$$

That window is the whole of the assumption, and the paper says so: under exotic emission scenarios it widens to $(-100, 1000)$ s, and the bound loosens with it by the same factor. The measurement is a lag and a distance; everything else in the number is what you were willing to assume about the burst.

Even so, fifteen orders of magnitude is enough to kill whole classes of modified gravity in one line of division.

What that number *removes*.

A great many attempts to explain the accelerated expansion without a cosmological constant add a scalar field coupled to curvature. In most of them the extra field changes the propagation of tensor modes, and gravitational waves travel at a speed that differs from c by something of order the departure from general relativity today.

Those models were being compared against cosmological data at the per-cent level. This measurement holds the same quantity to about one part in 10^{15} .

Ezquiaga and Zumalacárregui went through the surviving Horndeski classes one at a time after this result and removed most of them (PRL **119**, 251304). This is the one place in this course where a gravitational-wave measurement lands directly on the cosmology.

Tides in the last *orbits*.

Each star is deformed by its companion's tidal field, the deformation carries energy, and the energy shows up in the phase — at fifth post-Newtonian order, so it only matters in the last orbits, which is exactly where a neutron-star binary spends its cycles.

The measurable combination is a mass-weighted average of the two deformabilities. A stiffer equation of state gives larger stars, larger deformability and a phase that runs away from the point-particle prediction sooner. The data bound it from above, which bounds the radius from above, which excludes the stiffest candidates.

What is being constrained is matter at several times nuclear density. No laboratory reaches it, and the constraint arrived from an object 40 Mpc away. Which equations of state does that bound actually exclude, and at what confidence?

A distance without a *ladder*.

Day 1: the amplitude goes like $1/d$, and the chirp fixes the mass, so the amplitude measures the **luminosity distance** on its own. No rungs, no calibration, nothing borrowed from another distance indicator.

The counterpart gives the host galaxy, the host gives the redshift, and distance against redshift is H_0 .

WHERE THE WIDTH COMES FROM

Amplitude and inclination are degenerate: a nearer binary seen edge-on looks like a farther one seen face-on. That degeneracy, not the noise, sets the error on a single siren.

The host's peculiar velocity is the second term, and at 40 Mpc it is not negligible.

An amplitude is a distance and the counterpart is a redshift, so this is H_0 with no distance ladder under it. How much of the width is the distance–inclination degeneracy, and how much is the host's peculiar velocity?

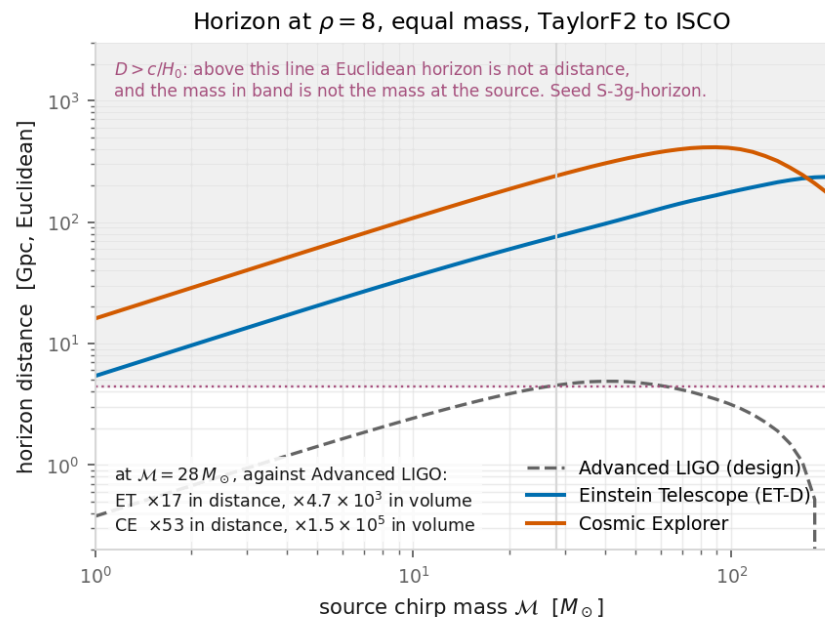
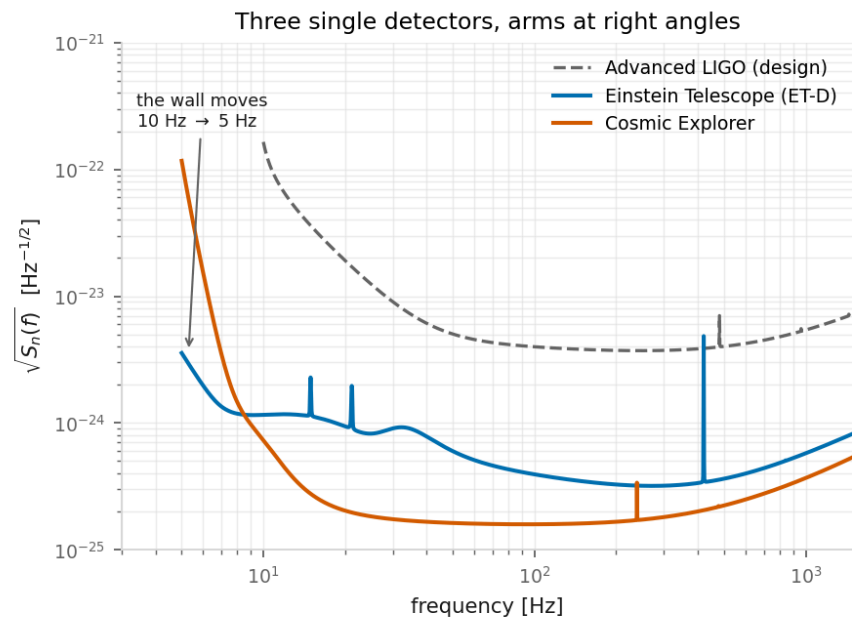


SECTION III

Louder and lower

A factor of ten in *sensitivity*.

Day 2 gave the whole argument: at fixed source $\rho \propto 1/D$, so a threshold defines a distance, and Euclidean volume goes as the cube of it — $d \ln V / d \ln \rho_{\text{thr}} = -3$. Ten times quieter is ten times further and a thousand times the volume. Against Advanced LIGO at design, the Einstein Telescope is a factor of 17 in distance and Cosmic Explorer a factor of 53. Both also move the low-frequency wall from 10 Hz to 5 Hz, which alone doubles the heaviest binary in band at all — from 440 to 879 solar masses.



The millihertz *band*.

Arm length sets the band. Kilometres put you at hundreds of hertz; millions of kilometres put you at millihertz, and that is a different sky.

WHAT LIVES THERE

- massive black-hole binaries, at the masses day 4 was counting;
- galactic white-dwarf binaries, so many that they are a foreground;
- small bodies spiralling into massive ones, hundreds of thousands of cycles deep in the strong field.

THE SAME BINARY, TWICE

A stellar-mass binary passes through the millihertz band years before it enters this one. Measure the phase there and you know when and where the merger will happen, long in advance.

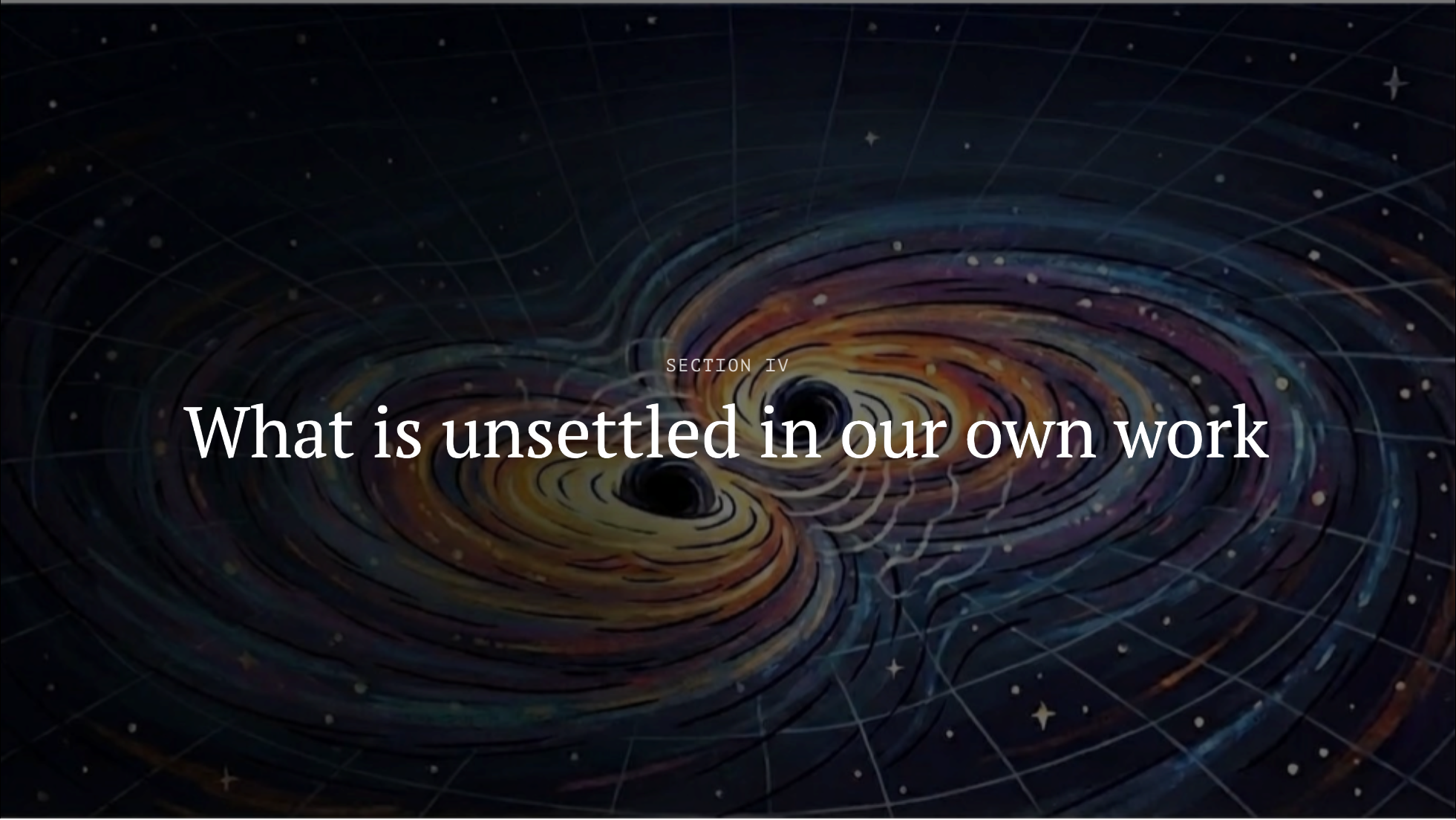
Which is a rare thing in astronomy: a prediction with a date on it. How well is the merger time pinned by the millihertz phase alone?

A background from the early *universe*.

Assume some process made gravitational waves in the early universe, when the temperature was T . Nothing has scattered them since — that is the whole reason to want them, and it is also why they are hard.

The frequency today follows from the horizon size then and the expansion since, so a band is a temperature. Ground-based frequencies correspond to the electroweak scale and above; the nanohertz band of day 4 sits far lower. A detection at any of them is a measurement of physics no accelerator will reach.

Day 1 named four classes of source and this week followed two of them. The other two — continuous waves and stochastic backgrounds — are still open to you, and the estimates you need are the ones you already made.

The background of the slide is a deep space illustration. It features two prominent spiral galaxies, one on the left and one on the right, rendered in vibrant shades of orange, yellow, and blue. These galaxies are set against a dark, star-filled sky. A faint, light-colored grid of lines, resembling a celestial map or a coordinate system, is overlaid on the entire scene. The overall aesthetic is artistic and cosmic.

SECTION IV

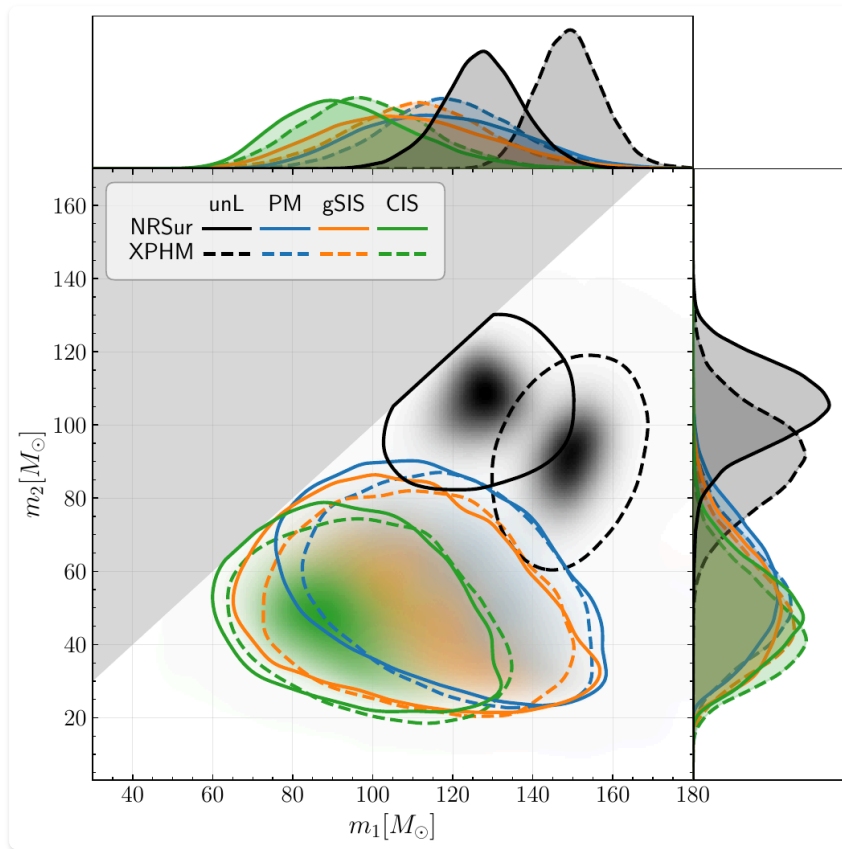
What is unsettled in our own work

The gap above the *heaviest*.

Stars in a window of helium-core mass become unstable to electron-positron pair production, blow themselves apart, and leave nothing behind. So the black-hole mass function should have a hole in it, somewhere above sixty solar masses and below about a hundred and thirty.

GW231123's components sit in that hole, or above it, depending on which waveform model and which interpretation you use.

Two ways out, and they are testable against each other. Either the components are themselves the products of earlier mergers, or the event is lensed and the true masses are lower.



Borrowed: Cheung et al., the GW231123 lensing release (Zenodo 16279016). Black: unlensed. Colours: three lens models, all of which move the masses down

The nanohertz *amplitude*.

Day 4 ended here, and it is worth restating as a problem rather than a conclusion. From a census of local supermassive black holes you predict the nanohertz background with no free parameters, and the prediction comes out a factor of a few low in black-hole mass density. No merger history closes that, because merging conserves mass.

1. **A heavier bright end** — the black holes carrying the signal are heavier than the census says, and there should be holes in nearby galaxies where they sit.
2. **A systematic in the census** — the relation, its scatter, or the aperture correction — and how far can the census move before the gap closes?
3. **Something that is not black-hole binaries** — and then the spectral shape has to match.

Each route predicts a different spectrum, and the measurement already has spectral information. And the question under all three: what is a single measured amplitude a statement about, when the median of the distribution sits below its mean?

Lensing, and the *Morse* phase.

A wave passing a mass can reach us along more than one path. In optics you see several images and compare their brightness. Here the signal is coherent, so the images add as amplitudes, and each one arrives with an extra phase that depends only on the type of stationary point it came through — not on frequency.

Those phases take discrete values, which makes them a signature rather than a nuisance parameter. When the images are not resolved in time they superpose into one distorted waveform, and the distortion is what you fit.

The GW231123 release is a complete public example: paper source, code, a locked environment and a map from every figure to the script that made it. It was published after any current model's training data ends, which makes it the hardest possible thing to hand an agent.

What is *open*.

Five days, and the honest summary is that the course opened more than it closed. These are the ones I would take, and none of them is settled.

WHERE THE PHYSICS RUNS OUT

Two modes, or one? The remnant is over-determined by any two, and a disagreement is a no-hair violation. Nobody has the second mode cleanly yet.

What the tidal bound excludes. It constrains a combination, not either star, and the translation to a radius is where the argument is.

The gap above the heaviest. Pair instability should empty a band, and GW231123 sits in or above it — unless it is lensed.

WHERE OURS DOES

The PTA amplitude. The measurement is above what the local census predicts. Heavier holes, a systematic in the census, or something that is not black-hole binaries — and the spectrum has to come out right either way.

What an anisotropy limit is a statement about. We showed one that reproduces its own prior. There will be others.

The timing array nobody here simulated. N pulsars, a common red process, and the question of how many years before a quadrupole is distinguishable from a monopole.

Every one of these is a week's work with what you now have on your machine, and for most of them nobody knows the answer. That is the point of ending here rather than on a summary.